

Skills Summary

Finding a Missing Number

Use this reference while completing your worksheet or when studying.

See the parts and the whole

Rule: In $\square + \triangle = \text{whole}$, the two parts go together to make the whole. If one part is hiding, the whole and the other part can find it.

Example: The whole is 11 and one part is 4. Find the other part.

Step 1. Put what we know in a part-whole box: whole 11 on top, part 4 below — the empty spot is what we want.

Step 2. 4 and how many more make 11? 4 and 7 make 11.

$$\Rightarrow 4 + 7 = 11$$

Try it: The whole is 12 and one part is 4. Find the other part.

check: 8

Count on to the total

Rule: Start at the part you know. Count up to the whole. The number of jumps is the missing number.

Example: $9 + \square = 13$

Step 1. We know where to start and where to land — count on from 9 up to 13 and keep track of the jumps.

Step 2. $9 \rightarrow 10, 11, 12, 13$. That is 4 jumps.

$$\Rightarrow 9 + 4 = 13$$

Try it: $8 + \square = 13$

check: 5

Use subtraction to find and check

Rule: Addition and subtraction are in the same fact family. The missing part is whole — known part, and adding back checks your answer.

Example: $\square + 5 = 12$

Step 1. The missing part comes first, so counting on is hard — subtract the known part from the whole instead.

Step 2. $12 - 5 = 7$. Check: $7 + 5 = 12$. It works!

$$\Rightarrow 7 + 5 = 12$$

Try it: $\square + 6 = 13$

check: 7

(!) Watch out: Do not add the two numbers you can see! In $\square + 6 = 13$, the answer is NOT 19. The whole is already 13 — the missing number is a part, so it must be smaller than the whole.